

Feeling the Direction: Understanding Affective Responses to Tactile Motion via a Wearable Pneumatic Interface

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Abstract

We present a soft, wearable pneumatic interface that conveys emotional meaning through directional tactile motion. By designing six dynamic tactile patterns and evaluating them in a user study (N=15), we explore how motion direction and speed shape emotional perception. Our findings suggest that directional tactile cues are recognizable and emotionally expressive, offering a foundation for designing affective feedback in intelligent soft wearables for communication, mindfulness, and human-agent interaction.

CCS Concepts

• Human-centered computing → Haptic devices.

Keywords

Wearable Tactile Interface, Tactile Motion, Social Communication, Pneumatic Haptic Feedback

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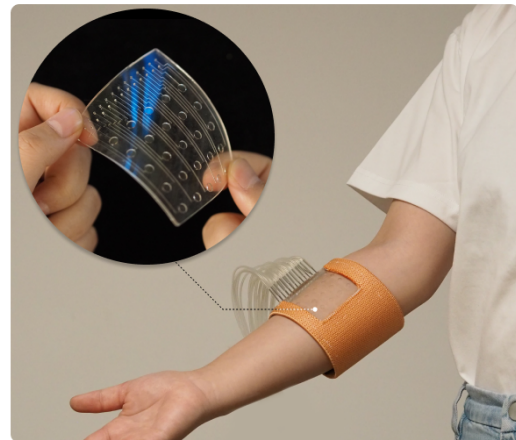


Figure 1: Wearable Soft Pneumatic Tactile Interface.

1 Introduction

Touch plays a vital role in human perception and emotional communication. Beyond basic sensing, tactile interaction can convey complex social signals such as gratitude, reassurance, or even anger [4, 10]. Neurophysiological studies have shown that tactile stimuli activate emotional regions of the brain, such as the insular and orbital frontal cortex, and can induce pleasant sensations via CT fibers [3, 18].

With the rise of remote communication, tactile feedback technologies have gained attention for supporting emotional expression beyond audio and visual channels [13]. Wearable haptic devices, particularly those using pneumatic actuation, are promising due to their comfort, expressiveness, and compatibility with the human body [2, 14, 17]. However, designing intuitive emotional tactile patterns remains a key challenge, especially when using low-resolution, sparse actuators.

Most existing studies have focused on tactile parameters such as intensity, duration, or temporal rhythm [5, 9, 15]. Yet the emotional

effects of *motion direction*, which means how tactile sensations dynamically move across the skin, remain underexplored. Previous studies have demonstrated that simulated light stroking stimulation on the forearm, which is sensitive to CT fibers, can induce pleasant sensations, but there have been limited studies on the emotional effects of tactile motion in other directions [11, 12]. We hypothesize that different directions of dynamic tactile motion can evoke distinct affective responses.

In this work, we explore how directional tactile motion influences emotional perception. We designed a wearable micro-pneumatic interface (see Figure 1) capable of producing programmable motion trajectories (e.g., horizontal, vertical, radial and circular), and conducted a user study to evaluate their recognizability, emotional valence-arousal attributes, and associated usage scenarios.

Our findings suggest that motion direction is a key factor in affective tactile design. Vertical motions were often perceived as supportive, while circular motions tended to evoke uncertainty. Radial expansion was associated with pleasant and comforting sensations, whereas radial contraction elicited different impressions depending on speed—appearing comforting when slow and tense when fast. These observations indicate that directional tactile motion can carry distinct emotional meanings. This work provides empirical evidence and actionable guidance for designing emotional haptic feedback based on motion direction.

2 System Overview and Tactile Motion Patterns

We developed a wearable pneumatic haptic interface capable of delivering programmable directional motion patterns across the forearm. The system comprises three components: (1) a flexible forearm strap, (2) a soft actuator array embedded in a microfluidic chip, and (3) a pneumatic control system.

Wearable Interface Design. The interface features a 5×5 array of independently addressable pneumatic actuators integrated into a soft PDMS microfluidic chip [8]. Each actuator consists of a cylindrical chamber that inflates when air is supplied, forming a temporary protrusion (air bubble) on the surface. The chip is enclosed within an elastic strap, ensuring consistent skin contact while maintaining comfort and wearability [17]. The deformable membrane directly touches the user’s skin, while transparent acrylic film and Velcro straps secure the interface for varying forearm sizes.

Pneumatic Control System. The actuators are driven by a pressure-regulated pneumatic system comprising a variable-frequency air compressor, a microfluidic pressure controller (FluidicLab¹), and two 16-channel solenoid valve array (two-position, 3-way, 50 Hz switching). The controller maintains stable airflow with <0.02% pressure fluctuation and a fast response time (<10 ms). The system operates at 350 mbar, balancing tactile clarity and safety. Control automation is achieved through FluidicLab Suite, which supports real-time adjustments, waveform generation, and programmable actuation sequences.

Directional Tactile Motion Patterns. To explore how motion direction influences affective interpretation, we designed six dynamic tactile patterns (see Figure 2), grouped into two motion types based on actuator activation trajectories:

- **Linear Motions:**

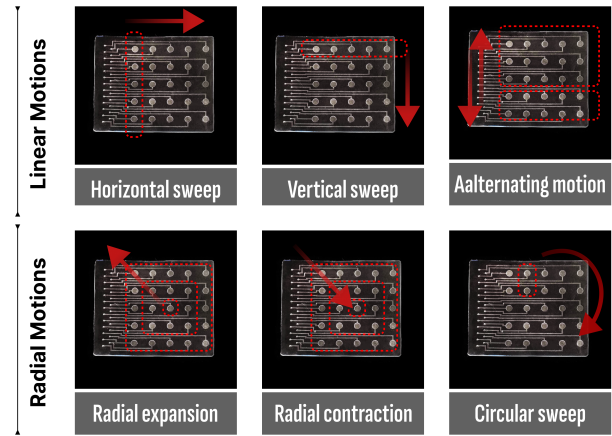


Figure 2: Directional tactile motion patterns.

- *Horizontal Sweep*: a lateral motion across the arm
- *Vertical Sweep*: a longitudinal motion along the arm (elbow to wrist)
- *Alternating Motion*: back-and-forth lateral shifts
- **Radial Motions:**
 - *Radial Expansion*: outward motion from the center to the periphery
 - *Radial Contraction*: inward motion from edges toward the center
 - *Circular Sweep*: circular motion along the actuator array perimeter

Each pattern was implemented in two temporal variants: slow and fast, both falling within the CT-optimal range (1–10 cm/s) as identified in prior research [11]. These speeds were selected through pilot testing to ensure perceptual clarity and avoid discomfort. The resulting motion patterns simulate natural affective gestures such as stroking, pressing, enclosing, or pacing, allowing us to examine how different motion directions influence emotional interpretation.

This classification allowed us to systematically assess how users perceive different types of motion in terms of emotional valence, arousal, and associated social meaning.

3 User Study and Key Findings

To evaluate how different tactile motion directions influence emotional perception, we conducted a user study focusing on pattern recognizability, affective interpretation, and usage scenario imagination.

Participants and Procedure. We recruited 15 participants (6 male, 9 female; mean age = 25.2, SD = 4.3) with no prior experience in tactile interfaces. Each session lasted approximately 40 minutes and consisted of three phases:

- *Familiarization*: Participants explored all six tactile patterns (in both slow and fast modes) while wearing the interface. Researchers provided visual cues and verbal descriptions to ensure consistent understanding.
- *Recognition Task*: Participants completed 30 randomized trials (6 patterns × 5 repetitions) and were asked to identify

¹<https://en.fluidiclab.com/products/fluid-control-systems/>

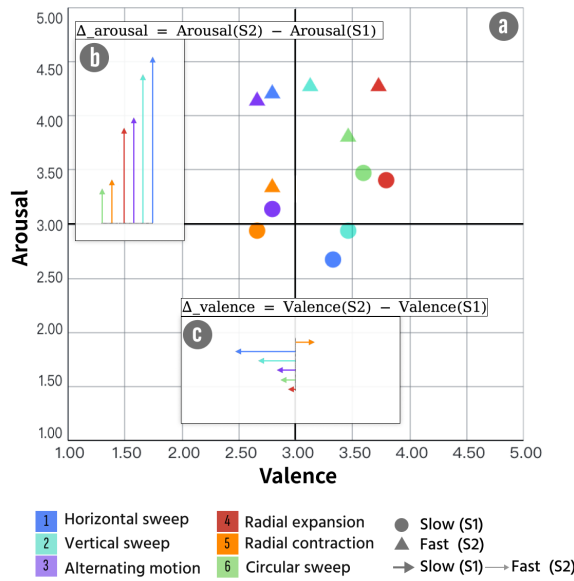


Figure 3: Valence and arousal results from directional tactile motion patterns. (a) Scatter plot of responses at slow (S1) and fast (S2) speeds. (b) Arousal changes (Δ arousal). (c) Valence changes (Δ valence).

which motion pattern they felt. Accuracy was used to assess distinctiveness.

- **Affective Labeling and Interview:** For each pattern, participants rated perceived emotions using a 2D valence-arousal grid and selected descriptors from a 10-label radar chart. Semi-structured interviews followed, exploring emotional impressions, metaphorical associations, and envisioned use cases.

Recognition Results. The average recognition accuracy across all patterns was 88.89%. Among the linear motions, *Horizontal Sweep* and *Vertical Sweep* were highly distinguishable (97.33% and 93.33%, respectively). Within radial motions, *Circular Sweep* achieved perfect recognition (100%), while *Radial Contraction* had the lowest accuracy (9.33%) due to its similarity with alternating motion in fast mode.

Emotional Perception of Motion Direction. Valence-arousal plots (Figure 3) revealed clear emotional trends tied to motion direction. We also identified the top three emotional attributes for each pattern at different speeds, which were summarized in the Table 1:

- *Vertical Sweep (Slow)* was associated with comfort, encouragement, and trust (high valence, low arousal).
- *Horizontal Sweep (Fast)* evoked tension and urgency (moderate valence, high arousal).
- *Radial Expansion* was perceived as uplifting or releasing (e.g., “like exhaling” or “relief”), while *Radial Contraction* suggested inward pressure or hesitation (e.g., “tightening grip” or “uncertainty”).
- *Circular Sweep* yielded mixed emotions such as “uncertainty,” “wandering,” or “anticipation,” especially in slow mode.

Speed significantly modulated affective interpretation: slower patterns were linked to positive emotions (e.g., comfort, calm), while faster patterns were interpreted as intense, alerting, or stressful.

Multimodal Associations. Participants often linked motion patterns to cross-modal impressions: - *Visual*: “rippling water,” “radar scan” - *Auditory*: “heartbeat,” “footsteps,” “gentle knock” - *Kinesthetic*: “stroking,” “pacing,” “hugging or releasing”

Usage Scenarios. Participants proposed a variety of emotional and situational applications: - *Vertical Sweep*: remote reassurance from a loved one; - *Radial Contraction*: waiting or tension feedback in messaging; - *Circular Sweep*: representing hesitation or “someone is typing...”; - *Alternating Motion*: decision-making support or internal conflict cues.

These results suggest that directional tactile motion carries emotional meaning both through physical cues and metaphorical embodiment.

4 Design Implications and Takeaways

Our findings highlight that directional tactile motion is not only perceptible but also carries distinct emotional connotations. This section summarizes key implications for affective haptic design, focusing on how motion direction, speed, and user perception contribute to emotional expressiveness.

Direction as an Emotional Cue. Motion direction emerged as a powerful dimension in shaping affective interpretation. *Vertical Sweep* was consistently associated with comforting or supportive gestures, aligning with natural upward strokes (e.g., petting or caressing) [12, 19]. In contrast, *Horizontal Sweep*, especially at higher speeds, was linked to urgency or confrontation. *Radial Expansion* conveyed openness or release, while *Radial Contraction* was interpreted as internalized tension or restraint. These directional cues resonate with embodied metaphors of emotional experience [1], such as “opening up” or “shrinking back.” In addition, our user study revealed that the emotional perception of tactile motion is influenced by the reference coordinate system. Specifically, whether users interpret motion direction based on a somatosensory (body-centered) or external (world-centered) frame of reference [16] can affect their reported valence and arousal. This finding suggests that designers of affective haptic systems may need to consider dynamic or context-sensitive alignment of motion direction relative to different coordinate systems.

Speed Modulates Intensity and Valence. The temporal aspect of motion had a clear impact: slower patterns were perceived as calming and positive, while faster ones were associated with high arousal and occasional discomfort. Designers should carefully balance motion speed to match intended emotional outcomes — e.g., slow expansion for reassurance, fast contraction for urgency or alert.

Motion Embodiment and Multimodal Associations. Participants frequently described tactile motion using visual, auditory, or bodily metaphors, such as “like a heartbeat,” “like someone pacing,” or “like ripples in water.” These multimodal associations suggest that directional haptics can tap into embodied memory and metaphor [6], enabling richer interaction design when combined with other sensory cues.

Table 1: Top three emotional attributes of directional tactile motion patterns.

2*Tactile Pattern	First Attribute		Second Attribute		Third Attribute	
	S1	S2	S1	S2	S1	S2
Pattern 1: Horizontal sweep	Comforting	Tense	Supportive	Pleasant	Pleasant	Surprised
Pattern 2: Vertical sweep	Supportive	Confident	Comforting	Pleasant	Surprised	Tense
Pattern 3: Alternating motion	Uncertain	Tense	Comforting	Uncertain	Confident	Supportive/ Confident/ Surprised
Pattern 4: Radial expansion	Comforting	Pleasant	Pleasant	Surprised	Confident	Confident
Pattern 5: Radial contraction	Comforting	Tense	Sad	Confident	Tense	Pleasant
Pattern 6: Circular sweep	Uncertain	Pleasant	Pleasant	Tense	Comforting	Surprised

Radial Motions Offer Emotional Subtleties. Compared to linear motions, radial patterns (expansion, contraction, circular) evoked more nuanced or ambiguous emotions — such as anticipation, uncertainty, or inner conflict. These are valuable in designing ambient feedback for subtle social or emotional states (e.g., “waiting for reply,” “processing,” or “thinking...”). Unlike discrete vibration signals, radial tactile motions afford continuous transitions and spatial metaphor.

5 Applications and Future Potentials

Based on participant feedback and our own design exploration, we envision several promising directions for integrating directional tactile motion into emotional interaction systems.

Remote Social and Emotional Signaling. Participants imagined using directional patterns to express presence, comfort, or subtle cues in remote communication. For example, a gentle *Vertical Sweep* could offer reassurance during a video call, while a *Circular Sweep* might indicate “please wait” or “I’m thinking.” These insights suggest the potential for low-disruption emotional signaling in collaborative, intimate, or asynchronous messaging contexts.

Affective Ambient Feedback. Patterns like *Radial Expansion* were seen as suitable for calming or motivational feedback during tasks, such as deep work, public speaking, or mindfulness practice. Integration with context-aware systems (e.g., biosensors, stress detectors) could enable real-time emotion-aligned tactile support.

Human-Agent Interaction. Directional tactile patterns could be used by embodied agents (robots, avatars) to express affective states non-verbally — for instance, a comforting *Expansion* pattern to acknowledge failure, or a *Contraction* pattern to express concern. These afford a tactile “body language” channel for future multi-agent systems.

Design Toolkits for Affective Haptic Vocabulary. Our tactile motion library may serve as a foundation for haptic designers to prototype and map emotional intentions through motion-based tactile expressions. With appropriate authoring tools, designers could compose patterns or create dynamic tactile metaphors tailored to specific contexts.

As this work progresses, we aim to explore personalization, cultural interpretation, and multi-modal integration, toward establishing directional tactile motion as a meaningful and expressive modality in affective computing.

6 Conclusion

This work investigates the affective potential of directional tactile motion delivered through a soft, wearable pneumatic interface. By designing and evaluating six dynamic tactile patterns, we demonstrate that motion direction and tempo can influence emotional perception in nuanced ways. Our findings suggest that directional tactile feedback may serve as a promising modality for emotional expression, particularly in contexts requiring subtle, embodied communication. This project contributes empirical insights and a structured design space for motion-based tactile feedback.

In future work, we aim to integrate sensing and adaptive control for real-time, context-aware affective interaction [7]. We also plan to examine the mechanisms behind tactile emotion decoding, including the role of multimodal associations and the influence of somatosensory versus external coordinate frames. Additional directions include exploring cultural generalizability and application in scenarios such as remote communication, ambient interfaces, and social robotics. Ultimately, we hope such interfaces may support well-being, social connection, and emotional self-awareness in intelligent soft wearable systems.

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